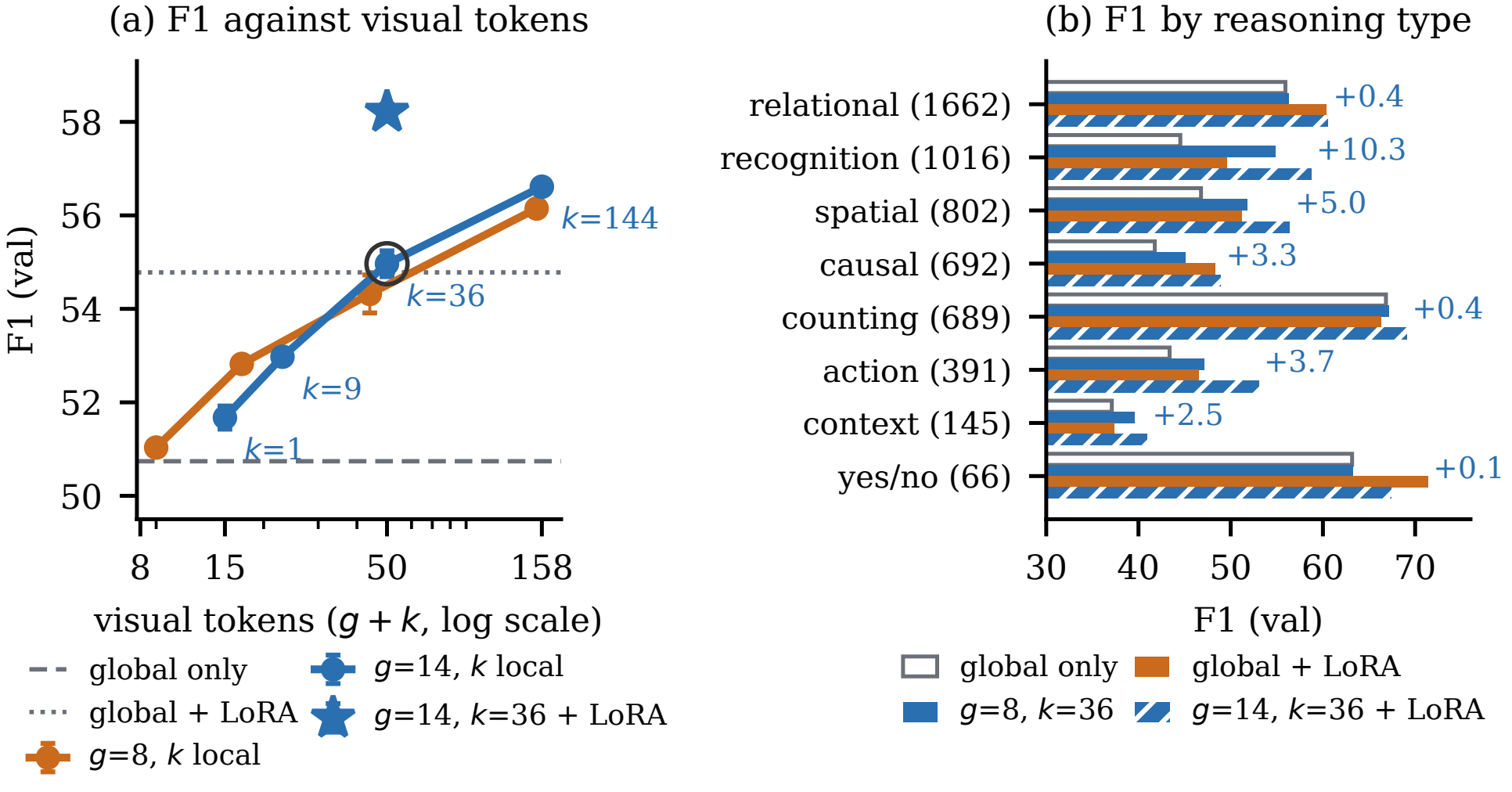
(a) F1 against visual tokens
F1 (val)
58
56
54
52
50
8
15
50
158
visual tokens ($g + k$, log scale)
k=144
k=36
k=9
k=1
global only
global + LoRA
g=8, k local
g=14, k local
g=14, k=36 + LoRA
(b) F1 by reasoning type
relational (1662)
recognition (1016)
spatial (802)
causal (692)
counting (689)
action (391)
context (145)
yes/no (66)
+0.4
+10.3
+5.0
+3.3
+0.4
+3.7
+2.5
+0.1
30
40
50
60
70
F1 (val)
global only
global + LoRA
g=8, k=36
g=14, k=36 + LoRA